

Clinical science

Survival of patients with recurrent uveal melanoma after treatment with radiation therapy

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ABSTRACT

Background/aims We evaluated a large cohort of patients treated for local recurrence of choroidal or ciliary body melanomas at the Massachusetts Eye and Ear (MEE) to quantify the risk of melanoma-related mortality associated with recurrence, independent of other risk factors.

Methods Patients treated with radiation therapy from 1982 to 2017 were identified through the Uveal Melanoma Registry at MEE. Competing risks regression was performed to investigate the risk of melanoma-related mortality associated with recurrence, treating recurrence as a time-varying covariate.

Results Of 4196 patients treated, 4043 patients remained recurrence-free and 153 patients experienced a recurrence (median follow-up: 9.9 years). Median time from initial treatment to recurrence was 30.5 months (range: 2.0–238.7). Seventy-nine (69.9%) patients with recurrences and 826 (37.9%) patients in the recurrence-free group died of metastatic uveal melanoma ($p<0.001$). Median time from initial treatment to melanoma-related death was 4.9 years (1.0–31.8) for patients who developed recurrences and 4.3 years (0.59–33.8) for patients who did not ($p=0.17$). Five-year and 10-year probabilities of melanoma-related mortality were 9.5% and 15.0%, respectively, in patients without local recurrences compared with 32.0% and 46.6% in patients with recurrences ($p<0.001$).

Conclusion These data confirm previous reports that local recurrence is associated with an increased risk of dying of melanoma and quantify the risk that can be attributed to local recurrence independent of other risk factors. This group of patients should be strongly considered for adjuvant therapies when available.

INTRODUCTION

Local recurrence of uveal melanoma after conservative treatments varies from 3% to 8%. Treatment options for recurrence include proton therapy, plaque therapy or enucleation. Factors associated with risk of developing a recurrence include tumour largest basal diameter,^{1–4} tumour thickness,⁵ ciliary body involvement of the tumour⁶ and age.^{4,5} Although it is known that an increased risk of dying from metastasis is associated with tumour recurrence^{2,3,7,8} in patients with uveal melanoma, few studies have quantified the long-term risk or determined if type of treatment for the recurrence affects survival. The Ophthalmic Oncology Task Force completed the most extensive study to date.⁹

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Local recurrence has been identified as a risk factor for melanoma-related mortality.

WHAT THIS STUDY ADDS

⇒ This study quantifies the long-term risk associated with local recurrence independent of other risk factors and demonstrates that the relative increase in risk of mortality from metastatic melanoma is greater for cases of recurrence in smaller tumours versus larger tumours.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The quantification of risk presented here will be useful in counselling patients with local recurrence. Additionally, the study findings suggest that tumours that recur may be inherently more malignant, indicating the need to further characterise such tumours and to develop adjuvant therapies for patients who experience recurrences.

In a cohort of 3809 patients with uveal melanoma, a strong association (HR: 6.28) between local recurrence and risk of metastasis was found.⁹ However, the median follow-up was short (3.7 years) and potential differences in outcomes based on re-treatment approach were not evaluated.

We evaluated a large cohort of patients treated for local recurrence of choroidal or ciliary body melanomas at the Massachusetts Eye and Ear (MEE) to quantify the risk of melanoma-related mortality associated with recurrence independent of other risk factors. Various factors potentially influencing risk of metastasis and mortality were investigated, including tumour size, recurrence type, and type of treatment for recurrence.

METHODS

Patients treated for choroidal or ciliary body melanomas between 1982 and 2017 were identified through the Uveal Melanoma Registry (UMR) at MEE.

The UMR contains clinical data including patient, tumour and treatment characteristics at initial examination as well as follow-up data regarding outcomes such as visual acuity, complications, eye loss and tumour recurrence. Patients are also



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Table 1 Recurrent tumour treatment

Treatment type*	N (%)
No treatment	5 (3.2)
Proton therapy†	77 (50.3)
Enucleation	64 (41.8)
Other‡	4 (2.6)

*Unknown if treated, n=3.
†Initial treatment: plaque RT (n=24), proton therapy (n=53).
‡Photocoagulation (n=2), plaque RT (n=1), unknown (n=1).
RT, radiotherapy.

followed to determine the incidence of metastatic melanoma and melanoma-related mortality. These data are ascertained through active surveillance, chart reviews and, for mortality data, reports from the National Death Index.

Specifically for the local recurrence endpoint, the date of the recurrence, date and type of treatment for the recurrence and the type of recurrence is ascertained and entered into the UMR. Type of recurrence is classified as follows: marginal recurrence, extrascleral extension, ring melanoma, uncontrolled growth and other type. Marginal recurrence is defined as growth that occurred at the borders of the treated tumour, and uncontrolled growth is defined as continued growth of the tumour in all dimensions, particularly thickness, after treatment.

For this analysis, we included patients with local recurrences who had been previously treated elsewhere with plaque brachytherapy and were referred to MEE for treatment of a recurrent tumour as well as patients who were treated with protons initially and then developed a recurrence while under observation. Patients treated previously by other means (eg, laser, transpupillary thermotherapy, iridectomy), typically for suspicious lesions or when the diagnosis was unclear, were excluded. Patients who had metastasis at the time of initial diagnosis of uveal melanoma were also excluded as were patients who died of uveal melanoma within 6 months of initial treatment, as it was likely that these patients had metastasis at the time of treatment.

Descriptive statistics were generated and differences by recurrence status were evaluated using Fisher's exact and Mann-Whitney tests as appropriate. A competing risks regression model was fit to patients with and without recurrences to investigate the risk of melanoma-related mortality associated with local

recurrence, treating recurrence as a time-varying covariate and other causes of death as a competing risk. Cumulative incidence of melanoma-related mortality was calculated to determine risk of death. In the subset of patients who had recurrences, Cox regression was used to assess if the type of treatment for the recurrence was associated with risk of death from melanoma.

RESULTS

Of 4196 patients treated during the study period and followed for approximately 12 years (mean: 12.0, median: 9.9), 4043 patients remained recurrence-free and 153 (3.6%) patients experienced a recurrence. Follow-up time was slightly longer in the recurrence-free group than in the recurrence group (median: 9.9 years vs 8.5 years, $p=0.17$). Twenty-four patients had received plaque radiotherapy elsewhere prior to recurrence. The remaining 129 patients were initially treated at MEE with protons and subsequently developed a recurrence; 53 (43.8%) were re-irradiated with protons while 64 (52.9%) underwent enucleation for the recurrence (table 1). An additional 10 patients who received proton therapy for recurrence subsequently underwent enucleation for continued growth. Recurrences could not be confirmed in 18 patients who were treated by enucleation outside MEE. Re-treatment status was unknown for 3 patients, and 5 patients were not treated for their recurrences which developed concurrently with metastatic disease. Median time from initial treatment to recurrence was 30.5 months (range: 2.0–238.7). Time to recurrence was similar in patients who were initially treated with plaque radiotherapy and those who received proton therapy (median: 34.1 months vs 29.3 months, $p=0.60$).

Baseline characteristics that are known risk factors for melanoma-associated mortality were found to be more common in patients who developed recurrences compared with those who remained recurrence-free (table 2). There were twice as many patients with T4 tumours (American Joint Committee on Cancer (AJCC) Staging Manual, 8th ed¹⁰) in the recurrence group than in the recurrence-free group. Conversely, one-third of patients who did not develop recurrences had T1 tumours while only 17% of patients with recurrences had T1 tumours. Differences in the tumour features that are used to determine staging (largest basal diameter, tumour thickness and ciliary body involvement of the tumour) are also shown in table 2. All

Table 2 Patient and tumour characteristics by recurrence status

	Recurrence n=153	Recurrence-free n=4043	P value
Median (range) follow-up time, years	8.4 (1.0–33.6)	9.9 (0.5–37.2)	0.17
Median (range) age at initial melanoma diagnosis	59.9 (22.0–87.1)	61.1 (10.2–94.9)	0.32
Median (range) LBD* (mm)	15.8 (3–25)	13.0 (3.5–26)	<0.0001
Median (range) thickness† (mm)	5.1 (1.0–14.5)	4.0 (0.6–17.1)	0.0001
Ciliary body involvement‡, n (%)	78 (51.0)	1001 (24.8)	<0.0001
AJCC classification§, n (%)			
T1	26 (17.1)	1319 (32.8)	<0.0001
T2	46 (30.3)	1310 (32.6)	
T3	51 (33.6)	1034 (25.7)	
T4	29 (19.1)	355 (8.8)	

*LBD was unavailable for 18 patients.

†Thickness was unavailable for 11 patients.

‡Information on ciliary body involvement was unavailable for one patient.

§AJCC classification was unavailable for 26 patients (25 in the recurrence-free group).

AJCC, American Joint Committee on Cancer; LBD, largest basal diameter.

Table 3 Types of recurrence by AJCC classification*, N (%)

	T1	T2	T3	T4
Marginal	9 (56.3)	23 (67.6)	26 (63.4)	13 (50.0)
Extrasceral	2 (12.5)	7 (20.6)	5 (12.2)	6 (23.1)
Ring melanoma	2 (12.5)	1 (2.9)	1 (2.4)	4 (15.4)
Uncontrolled	3 (18.8)	2 (5.9)	8 (19.5)	3 (11.5)
Other	0 (0.0)	1 (2.9)	1 (2.4)	0 (0.0)
p=0.38				
*Tumour recurrence type was unavailable in 36 patients. AJCC, American Joint Committee on Cancer.				

baseline characteristics were significantly different except for age at diagnosis; median age was approximately 60 years ($p=0.32$).

Marginal recurrences were the most common type ($n=71$, 46.4%) followed by extrasceral extensions ($n=20$, 13.1%) and uncontrolled tumours ($n=16$, 10.5%). Ring melanomas were uncommon ($n=8$, 5.2%) as were other types of tumour recurrences ($n=2$, 1.3%). Tumour recurrence type was unavailable in 36 patients (23.5%). There was no predilection for any recurrence type to occur in a particular tumour size class (table 3). Marginal recurrences accounted for 56.3% ($n=9$) of T1 tumours, 67.6% ($n=23$) of T2 tumours, 63.4% ($n=26$) of T3 tumours and 50.0% ($n=13$) of T4 tumours. Three (18.8%) patients had T1 tumours that were uncontrolled, while 2 (5.9%) patients with T2 tumours, 8 (19.5%) patients with T3 tumours and 3 (11.4%) patients with T4 tumours experienced uncontrolled growth ($p=0.38$).

Information on treatment for recurrence in addition to type of recurrence was available for 105 patients. Marginal recurrences were more likely to be re-treated with proton beam irradiation (PBI) ($n=38$, 59.4%) than all other types of recurrences for which enucleation was favoured ($n=26$, 63.4% enucleated) ($p=0.03$).

There were 2321 (55.3%) deaths in the cohort. Cause of death was unavailable in 32 cases; of patients with a known cause, 905 (39.5%) were melanoma-related. Over two-thirds ($n=79$, 69.9%) of patients with recurrences died of metastatic uveal melanoma, while 37.9% ($n=826$) in the recurrence-free group died of uveal melanoma ($p<0.001$). Median time from initial treatment to melanoma-related death was 4.9 years (1.0–31.8) for patients who developed recurrences and 4.3 years (0.59–33.8) for patients who did not ($p=0.17$). Excluding patients with concurrent metastasis and recurrence (date of metastasis before or within 1 month of recurrence date), survival varied from as little as 1 month after diagnosis of recurrence to as long as 16 years (median: 28.3 months).

Gene expression profiling (GEP; DecisionDx-UM, Castle Biosciences) was available for 74 patients in the cohort; 4 of these developed recurrences. Class 2 tumours accounted for 75% (3/4) of the tumours in the recurrence group compared with 28.6% (20/70) of the tumours that did not recur. Among the patients with recurrences, one patient with a class 2 tumour died of melanoma 5.7 years after initial treatment and 4.5 years after recurrence. The remaining 3 patients are alive approximately 4 years after initial treatment and 23.3 months after recurrence (mean follow-up times).

Of the 70 patients without recurrences, melanoma-associated deaths occurred exclusively in patients with class 2 tumours; 7 of 20 patients died in this group, all related to melanoma; mean survival time after initial treatment was 2.3 years. One patient with a class 1A tumour died with unknown cause of death and

Table 4 Competing risks regression evaluating the effect of local recurrence on melanoma-related mortality

Model 1 Recurrence present or absent					
Risk factor		Univariable		Multivariable	
		SHR (95% CI)	P value	SHR (95% CI)	P value
Recurrence	no recur (ref) vs recur	5.05 (3.99 to 6.38)	<0.001	3.87 (3.08 to 4.85)	<0.001
AJCC classification	T1 (ref)				
	T2	2.79 (2.17 to 3.57)	<0.001	2.75 (2.14 to 3.52)	<0.001
	T3	6.27 (4.95 to 7.93)	<0.001	6.08 (4.80 to 7.70)	<0.001
	T4	13.39 (10.41 to 17.23)	<0.001	12.59 (9.78 to 16.20)	<0.001
Age at treatment	1 year	1.002 (0.99 to 1.006)	0.33	1.003 (0.99 to 1.007)	0.15
Model 2 Recurrence type					
		SHR (95% CI)		SHR (95% CI)	
Recurrence type*	No recur (ref)				
	Marginal recurrence	5.76 (4.18 to 7.94)	<0.001	4.33 (3.17 to 5.94)	<0.001
	Other types	5.97 (4.13 to 8.63)	<0.001	3.51 (2.42 to 5.08)	<0.001
AJCC classification	T1 (ref)				
	T2	2.78 (2.16 to 3.56)	<0.001	2.73 (2.13 to 3.51)	<0.001
	T3	6.27 (4.95 to 7.95)	<0.001	6.04 (4.76 to 7.66)	<0.001
	T4	13.37 (10.37 to 17.23)	<0.001	12.51 (9.69 to 16.14)	<0.001
Age at treatment	1 year	1.002 (0.99 to 1.006)	0.32	1.003 (0.99 to 1.007)	0.16
*Type of recurrence was unknown in 36 patients (23.5%). SHR, subhazard ratio.					

one patient with a class 1B tumour died of a non-cancer cause. No further assessment was completed assessing GEP, recurrence and mortality because there was insufficient data for more robust analyses.

Competing risks regression revealed overall a higher than threefold greater risk of melanoma-related mortality associated with recurrence after adjusting for AJCC classification, age and the possibility of death from other causes (subhazard ratio: 3.87, 95% CI: 3.08 to 4.85, $p<0.001$) (table 4). Cumulative incidence of melanoma-related mortality at 5 and 10 years after initial diagnosis and treatment were 9.5% and 15.0%, respectively, in patients who did not experience local recurrence compared with 32.0% and 46.6% in patients with recurrence (figure 1). Examining results by AJCC tumour size categories, these differences persisted with higher rates of melanoma-related mortality in patients with recurrences for all tumour size groups (figure 2). The increased probability of death from metastatic disease associated with recurrence appears to be somewhat greater in smaller tumours than in larger tumours with an over threefold difference in cumulative incidence rates in patients with T1 (19.9% vs

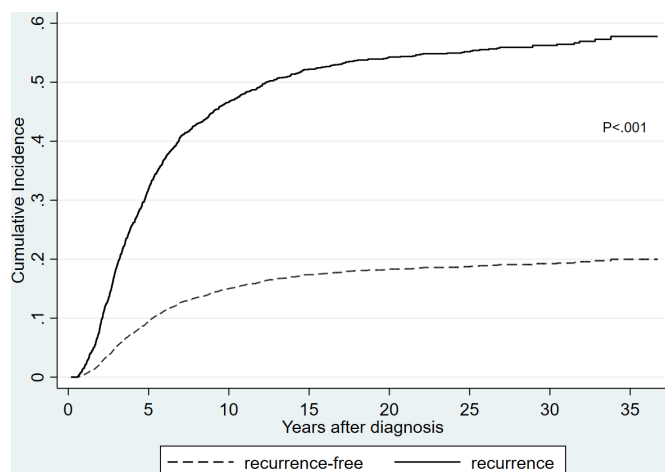


Figure 1 Cumulative incidence of uveal melanoma-related deaths in the presence of competing risks, by tumour recurrence status.

5.6%) and T2 (45.6% vs 14.6%) tumours at 10 years. The probability of death in patients with T3 tumours that recurred was approximately 2.5 times greater than that of patients without recurrences (cumulative incidence at 10 years: 74.0% vs 29.4%) but less than twofold greater in the T4 group (93.8% vs 51.4%).

We also evaluated whether type of recurrence was associated with risk of mortality. Differences in risk by tumour recurrence type were not observed: patients with marginal recurrences and other types of recurrences were at a similar increased risk of melanoma-related death compared with patients who were

recurrence-free (subhazard ratio (SHR): 4.33 (95% CI: 3.17 to 5.94) and 3.51 (95% CI: 2.42 to 5.08) for marginal recurrences and other types of recurrences, respectively) (table 4).

To determine if type of treatment influenced survival outcome, we compared patients with recurrences who were treated with proton therapy to those who underwent enucleation. Treatment type did not alter outcome. Similar melanoma-associated mortality rates were observed in both treatment groups; 10-year rates were 31.1% (95% CI: 21.9 to 43.1) and 55.6% (95% CI: 43.3 to 68.7) in the proton and enucleation groups, respectively (log rank $p=0.17$, adjusted for AJCC staging). There was a trend for recurrences in T3 and T4 tumours to be treated by enucleation; over half of patients with these tumours had their eyes removed. However, Cox regression did not reveal a higher risk of mortality in patients who received proton therapy for the recurrence after adjusting for tumour size (HR, protons vs enucleation: 1.26, 95% CI: 0.77 to 2.05; table 5).

DISCUSSION

Local recurrence is known to elevate risk of dying from uveal melanoma.^{2,9} Past studies evaluating tumour recurrence have reported an association with an increased risk of death.^{7,11–13} One study assessed the risk of death associated with recurrence after proton therapy and found 10-year rates of survival of 83.6% and 55.0% in recurrence-free and recurrence groups, respectively.² Similar findings were reported by the Ophthalmic Oncology Task Force⁹; 10-year metastasis-free survival for patients without recurrences was 82% while the rate was 62% for those with local recurrences.⁹ Results from these two studies are consistent with the overall 10-year cumulative incidence of deaths from uveal

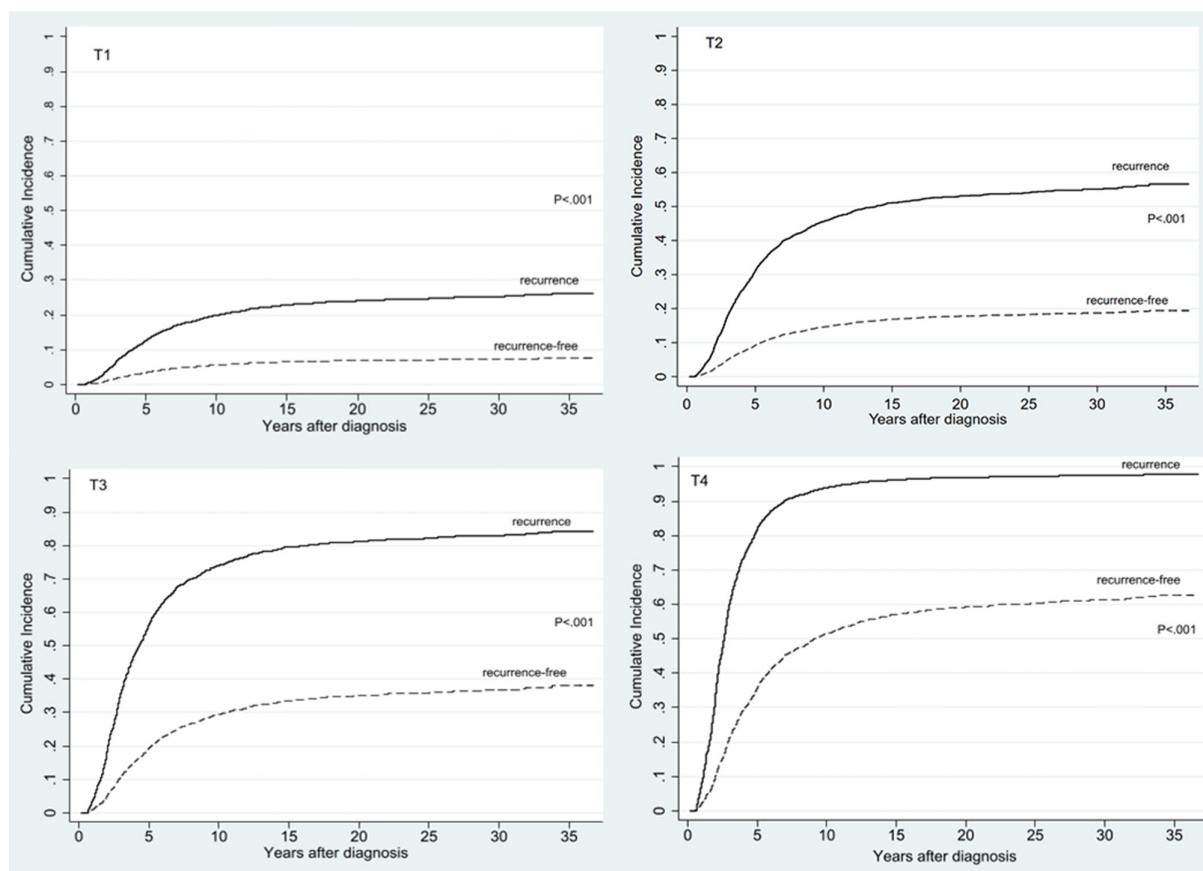


Figure 2 Cumulative incidence of uveal melanoma-related deaths in the presence of competing risks, by tumour recurrence status and AJCC tumour classification. Upper left: T1 tumours; upper right: T2 tumours, lower left: T3 tumours, lower right: T4 tumours.

Table 5 Cox regression evaluating the effect of re-treatment type on melanoma-related mortality in patients with recurrences

Risk factor		Univariable	P value	Bivariable	P value
		HR (95% CI)		HR (95% CI)	
Treatment for recurrence	PBI (ref) vs enucleation	1.66 (1.04 to 2.66)	0.03	1.26 (0.77 to 2.05)	0.36
AJCC classification	T1 (ref)				
	T2	2.82 (2.19 to 3.62)	<0.001	1.58 (0.62 to 4.07)	0.34
	T3	6.48 (5.11 to 8.22)	<0.001	2.36 (0.95 to 5.88)	0.06
	T4	14.23 (11.06 to 18.31)	<0.001	6.98 (2.76 to 17.63)	<0.001

AJCC, American Joint Committee on Cancer; PBI, proton beam irradiation.

melanoma found in our study (15.0% in recurrence-free patients and 46.6% in patients with recurrences).

In contrast to our findings, the Ophthalmic Oncology Task Force did not find an association between AJCC tumour size and recurrence, and the association between recurrence and metastasis in multivariable analysis was stronger in their study⁹ (HR=6.46 vs SHR=3.87 in our study). Differences in methods of analysis and the outcome of metastasis rather than melanoma-related mortality may account for these differences.

The cohort presented here represents the largest series of patients with local recurrence after radiation for uveal melanoma with the longest follow-up. These data further quantify the risk of melanoma-associated mortality incurred when initial primary tumour control is not achieved. Additionally, the size of this cohort allowed for more reliable evaluation of the increased mortality associated with local recurrence in different tumour size categories. The relative increase in mortality risk appears highest for those with the smallest tumours. The 10-year risk of death from metastatic melanoma for patients with T1 tumours was 5.6% in those without local recurrence compared with 19.9% in those with local recurrence. While the absolute risk of melanoma-related mortality was significantly higher in those with larger primary tumours, the relative increase in risk associated with a local recurrence was smaller at 1.8-fold for T4 tumours vs 3.6-fold for T1 tumours. These results underscore the importance of effective treatment for small tumours as one of the most impactful means to reduce melanoma-associated mortality. However, since this analysis was restricted to patients who experienced local recurrence after radiotherapy for melanoma, it cannot be assumed that the same increase in mortality would be seen after local treatment failure from other treatment modalities such as laser-based therapies.

GEP was unavailable for the majority of patients in this study, precluding exploration of associations between molecular risk factors, local recurrence and melanoma-related mortality. Recent findings of Bagger and colleagues¹⁴ demonstrated that local recurrence remained an independent risk factor for metastatic disease in patients whose tumours showed copy number alterations in chromosome 3 and 8q (HR=2.7). However, chromosome 3 and 8q status of the tumour did not influence the risk of local recurrence.

Our findings with respect to risk of mortality and type of recurrence differ in comparison to prior studies. Other studies with fewer cases and shorter follow-up observed a lower risk of metastasis and melanoma-related mortality with marginal recurrences vs other types of recurrences.^{2,15} However, the analysis presented here showed a similar increase in risk of death from metastatic melanoma for patients with marginal vs other types of recurrences. This result may be limited by lack of data on type of recurrence for 24% of the patients in this series, but the longer follow-up for our cohort likely also was a factor.

With respect to treatment for local recurrences, we found no differences in survival between those treated with proton irradiation and enucleation, validating previous findings from our site.¹⁶

CONCLUSIONS

These data confirm previous reports that local recurrence is associated with an increased risk of dying of melanoma and quantify the risk that can be attributed to local recurrence independent of other risk factors. This group of patients should be strongly considered for adjuvant therapies when available. The observation that the increased risk of mortality associated with recurrence remains significant after adjusting for tumour size suggests that those tumours that recur locally after radiation may be inherently more malignant. Further characterisation of tumour and host factors in cases of local recurrence is required.

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Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by the Mass General Brigham Institutional Review Board, conducted under Protocol 2019P000606, Epidemiologic Studies of Uveal Melanoma, and Protocol 2019P000285, the Uveal Melanoma Repository.

Provenance and peer review Not commissioned; externally peer reviewed.

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